

440.463 – Economics of Investments and Financial Management

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Introduction

We are junior traders at Columbus Capital preparing an NYSE trading strategy to increase return over a three-month investment period. Our objective is a portfolio which will achieve strong risk-adjusted returns in conjunction with reasonable diversification and risk management under changing market conditions. To construct the portfolio, we leverage macroeconomic and industry analysis to take 20 NYSE-listed stocks from sectors that are supported by positive momentum and solid fundamentals. Then we apply fundamental analysis, technical analysis, and portfolio optimization to rank these stocks using both quantitative and fundamental scoring models and eventually select 10 stocks for the final portfolio. We offer a moderately aggressive portfolio. We wish to get an equal-weighted benchmark based on the existing universe of 20 stocks. The above have the limited constraints which are all the stocks must be listed on NYSE, the portfolio should diversify across cyclical and defensive sectors, and every chosen stock should have the minimum weight for the portfolio, 5%.

Methodology

Step 1 - Opening an Account

Account name: JHUEIFMteamnew.

Step 2 - First Choice of 20 stocks on NYSE

Our final 20 stocks are ABBV, APD, ASX, BHP, CAT, COP, CVX, GE, HSY, JNJ, LIN, LLY, MMM, NEM, PG, RTX, SHEL, TSM, UNH, and XOM. These selected stocks cover six main sectors and several industries. The energy sector contains major oil and gas companies (including SHEL, COP, XOM, and CVX). The material sector includes companies such as industrial gases, mining, chemicals, and gold mining (including LIN, BHP, APD, and NEM). The semiconductor sector includes semiconductor manufacturing and foundry businesses like ASX and TSM. For defensive exposure, we include Healthcare companies (including JNJ, LLY, ABBV, and UNH) that cover pharmaceuticals, healthcare products, and managed healthcare. The Consumer Staples companies PG and HSY represent household products and packaged food. The industrial sector includes companies covering machinery, aerospace and defense, and industrial conglomerates CAT, RTX, GE, and MMM.

To reach this portfolio, we conduct industrial analysis including a top-down and bottom-up approach along with the macroeconomic factors that keep up with events happening in the real world. Global tensions stemming from the war in Ukraine and the Israeli-Palestinian conflict make the financial markets volatile and uncertain. As a result, demand for energy and defense industries is expected to grow, potentially pushing oil prices higher. For these reasons, we chose the Energy and Industrials sectors for stock selection. The Industrials sector also benefits

from increased infrastructure spending, supply chain shifts, and higher global defense budgets. This approach matches current market trends.

With top-down approach, we start by looking at the overall economy before picking stocks. We pay attention to factors such as interest rates, economic growth, inflation, the job market, consumer confidence, and global risks. Right now, the economy appears to be in a late expansion phase, which means there is more inflation and volatility. Because of this, we include both cyclical and defensive sectors. Energy, Materials, Semiconductors, and Industrials can help the portfolio grow, while Healthcare and Consumer Staples help manage risk. With bottom-up approach, we pick individual companies from these sectors. We look for firms that have strong positions in the market, stable operations, good growth prospects, and enough data for future analysis so we can make sure the stocks are not just in promising sectors but are also a good fit for our return and risk goals.

Our macroeconomic analysis uses three indicators: T10Y3M, initial claims, and consumer sentiment. The yield curve is gradually normalizing after the recession. The labor market is relatively stable. Consumer sentiment is slightly recovering. Based on these indicators, we believe the economy is in a late expansion phase. However, the market still faces slow economic growth, persistent inflation pressure, and increasing market volatility. Because of this, cyclical sectors may still perform well, but defensive sectors are also important for risk control.

Step 3 - Data Preparation

Details are in our Excel file.

Step 4 - Financial Statement Analysis

Our project excels table presents the fundamental analysis part of our scoring model. We use financial data from Yahoo Finance, such as ROE, Net Margin, Revenue Growth, Debt/Equity, P/E, EV/EBITDA, to analyze the operating, financial, and profitability conditions of the 20 companies.

Based on this, we standardize all variables using z-scores so that they can be compared across stocks. Finally, we compute the fundamental score (FundScore) using the following formula:

$$\text{FundScore} = 0.30 * z_{\text{ROE}} + 0.20 * z_{\text{NetMargin}} + 0.30 * z_{\text{Growth}} + 0.10 * z_{\text{Debt/Equity}} + 0.05 * z_{\text{P/E}} + 0.05 * z_{\text{EV/EBITDA}}$$

Companies with stronger profitability and growth will get higher scores, while companies with higher leverage and valuation will be penalized. The FundScore is used to evaluate the overall financial quality of each company and serves as a supporting factor in the stock selection process.

Step 5 - Optimization

We calculate performance indicators including annualized Sharpe ratio, alpha, and beta. These indicators measure risk-adjusted return, excess return, and market risk exposure. We standardize all variables using z-scores so that they can be compared across stocks. We compute the market-based score using the following formula:

$$\text{Score} = 0.7 * z_Sharpe + 0.3 * z_Alpha - \lambda * |z_Beta|$$

Stocks with higher Sharpe ratio and alpha receive higher scores, while stocks with higher absolute beta are penalized to control risk.

After that, we combine this performance score with the fundamental score (FundScore) from the previous part to obtain the final score (FinalScore) by taking the average, which is used to rank the stocks.

$$\text{Final Score} = 0.5 * \text{Score} + 0.5 * \text{FundScore}$$

The purpose of this step is to consider return, risk, and fundamental quality together in the selection process instead of relying on a single metric, and to make the decision more stable and balanced.

Among all the stocks, SHEL, ASX, COP, and LIN show strong performance based on historical market data. On the other hand, BHP, JNJ, CAT, and TSM show relatively stronger profitability, growth potential, or valuation conditions based on fundamental analysis.

However, the 10 stocks we select are not simply the top 10 stocks with the highest final scores. During the selection process, we also apply sector constraints, including selecting a maximum of two and a minimum of one stock from each sector, while maintaining a balance between offensive and defensive sectors.

Based on our strategy, the final 10 stocks selected for portfolio construction are SHEL, ASX, COP, LIN, BHP, JNJ, CAT, TSM, PG, and RTX.

After selecting the final 10 stocks, we apply the Markowitz Mean-Variance Optimization framework to determine the portfolio weights. We use 10 years of daily adjusted return data to estimate the expected return vector and covariance matrix of the selected stocks.

The optimization objective is to construct a maximum Sharpe ratio portfolio under several portfolio constraints. First, the sum of portfolio weights must equal 1. Second, short selling is not allowed, so all portfolio weights must remain non-negative. Third, we impose minimum weight constraints to avoid excessive concentration in a small number of stocks.

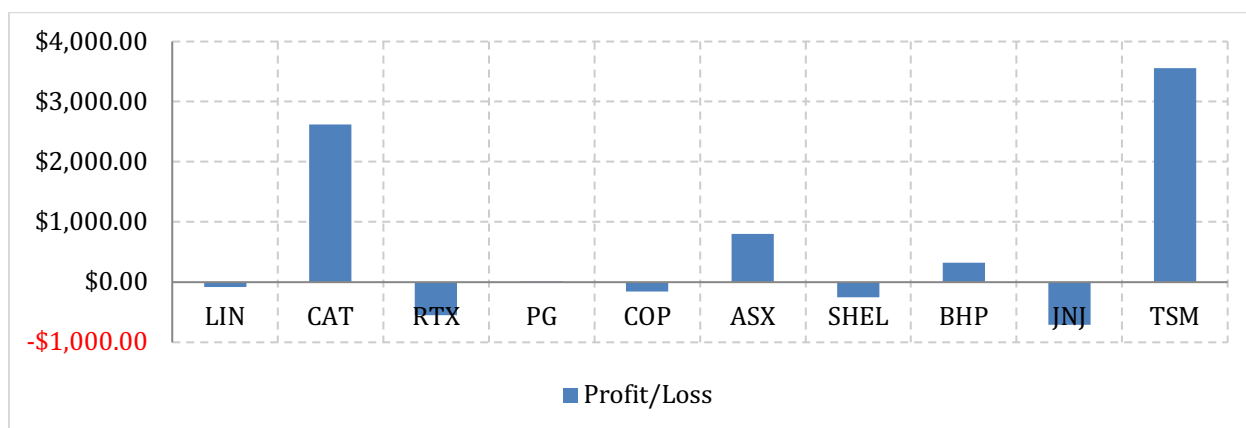
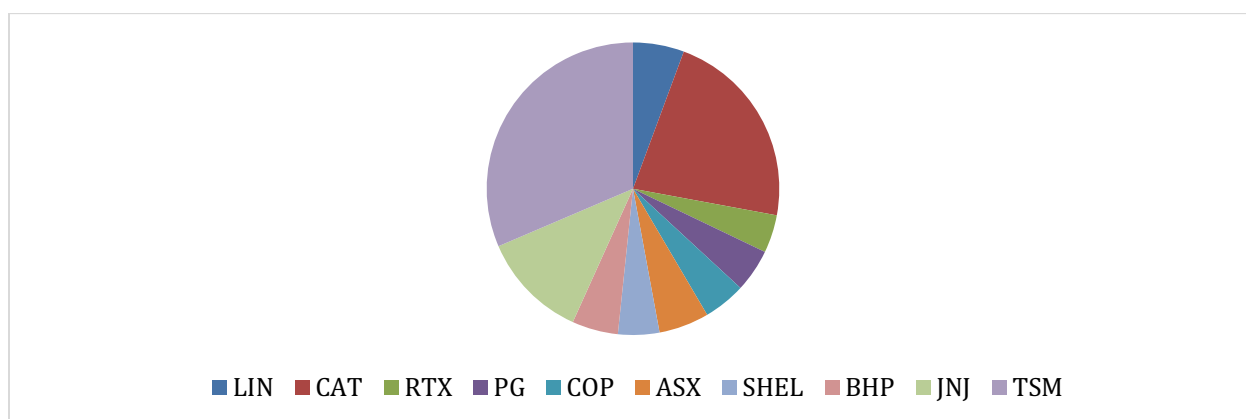
We also incorporate our portfolio preference into the optimization process. Semiconductor, Energy, and Industrials are treated as offensive sectors, while Healthcare and Consumer

Staples are treated as defensive sectors. Based on our moderate-aggressive strategy, we target approximately 70% allocation to offensive sectors and 30% allocation to defensive sectors.

The optimization is implemented in Python. Compared with equal-weight allocation, the optimized portfolio achieves a better balance between expected return and risk.

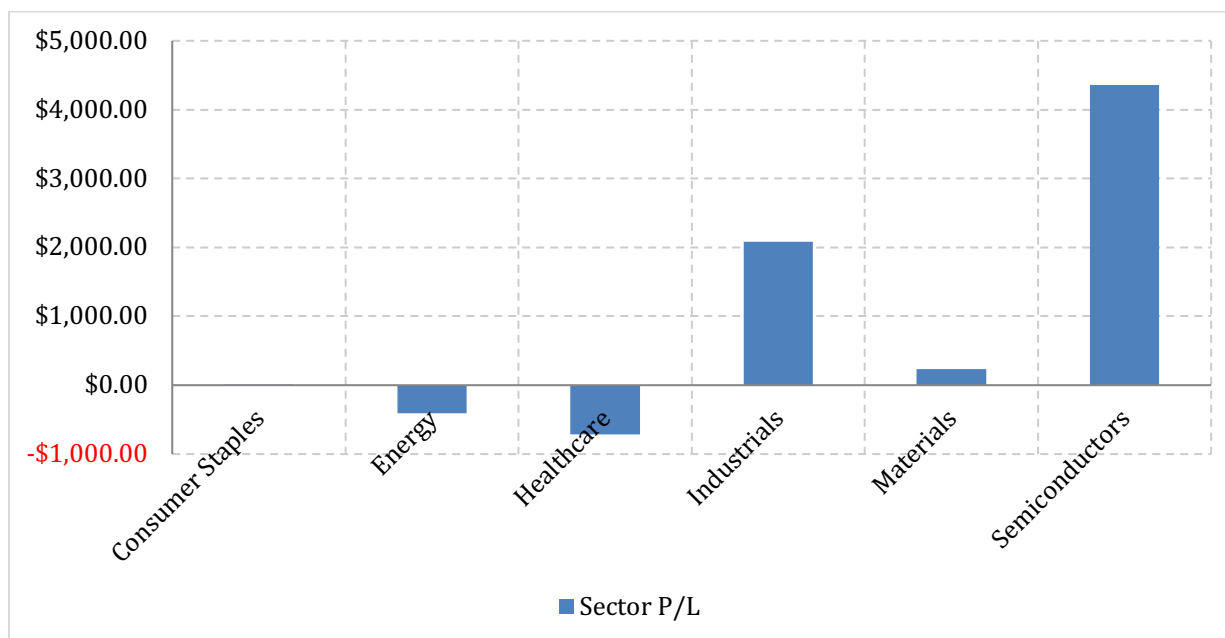
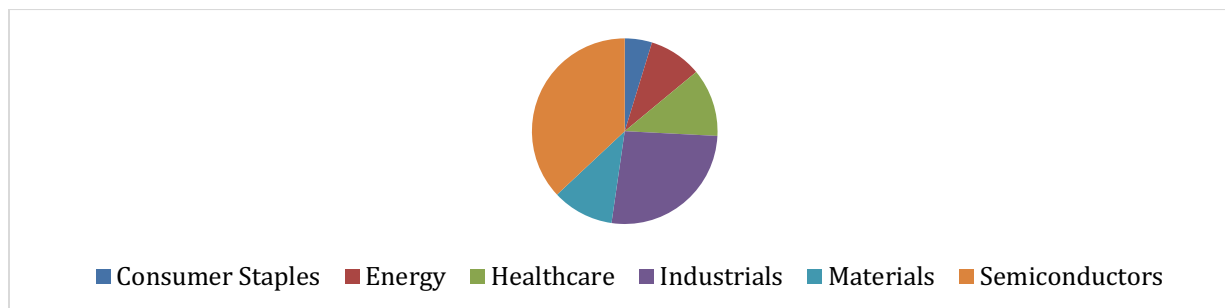
Open Positions

Symbol	Sector	Quantity	Price Paid	Last Price	Day's Change	Cost Basis	Market Value	Profit/Loss	% Return
LIN	Materials	12	\$499.93	\$493.16	-0.69	\$5,999.16	\$5,917.92	-\$81.24	-1.35%
CAT	Industrials	26	\$796.47	\$897.45	1.76	\$20,708.22	\$23,333.70	\$2,625.48	12.68%
RTX	Industrials	25	\$197.96	\$176.09	-0.69	\$4,949.00	\$4,402.25	-\$546.75	-11.05%
PG	Consumer Staples	34	\$146.16	\$146.42	0.36	\$4,969.44	\$4,978.28	\$8.84	0.18%
COP	Energy	43	\$117.55	\$113.87	-1.01	\$5,054.65	\$4,896.41	-\$158.24	-3.13%
ASX	Semiconductors	170	\$29.52	\$34.23	0.89	\$5,018.40	\$5,819.10	\$800.70	15.96%
SHEL	Energy	57	\$88.30	\$83.97	-0.27	\$5,033.10	\$4,786.29	-\$246.81	-4.90%
BHP	Materials	63	\$79.66	\$84.72	2.17	\$5,018.58	\$5,337.36	\$318.78	6.35%
JNJ	Healthcare	56	\$234.05	\$221.32	-1.19	\$13,106.80	\$12,393.92	-\$712.88	-5.44%
TSM	Semiconductors	80	\$367.24	\$411.68	-2.47	\$29,379.20	\$32,934.40	\$3,555.20	12.10%
Total Open Positions						\$99,236.55	\$104,799.63	\$5,563.08	5.61%



Sector Summary

Sector	Cost Basis	Market Value	Profit/Loss	Weight
Consumer Staples	\$4,969.44	\$4,978.28	\$8.84	4.75%
Energy	\$10,087.75	\$9,682.70	-\$405.05	9.24%
Healthcare	\$13,106.80	\$12,393.92	-\$712.88	11.83%
Industrials	\$25,657.22	\$27,735.95	\$2,078.73	26.47%
Materials	\$11,017.74	\$11,255.28	\$237.54	10.74%
Semiconductors	\$34,397.60	\$38,753.50	\$4,355.90	36.98%
Total	\$99,236.55	\$104,799.63	\$5,563.08	100.00%



Step 6 - Technical Analysis

According to technical analysis, the US stock market as a whole has rebounded significantly from its low point at the end of March, around April 20th. The data show that the S&P 500 experienced a sustained recovery after the end of March. By around April 20th, it had achieved a relatively strong rebound performance. Some momentum and growth assets have also started to outperform the broader market, reflecting the capital that has flowed back into risky assets and the market's short-term trend shifting gradually from defensive to aggressive.

Then we compared main industries' ETF performance, which is used to determine the flow of market funds and the momentum of industries. The results showed that the materials, energy,

semiconductors, and consumer staples sectors performed relatively better, indicating that these industries have garnered more market attention and momentum. Based on this point, we gave priority to selecting stocks from these strong sectors rather than distributing them average to all industries.

Third, at the individual stock level, we focused on the stock price, whether it is in a relatively upward trend and whether it is performing stronger than the stock market and industry ETF. If the stock had a strong performance and it could recover quickly after a pullback, we believe that it would have better short-term momentum and be more suitable for inclusion in the portfolio. If the stock price is weaker than the market or industry, it will be deprioritized, even if it has good fundamentals.

Overall, our technical analysis focuses on determining the market trends and timing of entry. The rationale for buying on April 20th was the market showed signs of stabilization after a correction, strong sectors maintained relative momentum, and some leading stocks outperformed the market. Therefore, at this point in time, not only can the risk of premature decline be avoided, but also there is an opportunity to participate in the subsequent market rebound.

Converting Portfolio Weights into Actual Trades

Once we had our optimal portfolio weights, we needed to figure out how many shares to buy for each stock. We started with \$100,000 in our portfolio. For each stock, we multiplied this amount by the target weight to get the dollar amount we wanted to invest.

Target Dollar Amount = Portfolio Value x Target Weight

Next, we divided that dollar amount by the stock's price on the day we bought it to get the number of shares.

Shares to Buy = Target Dollar Amount / Purchase Price

Since we can only buy whole shares, we rounded down to the nearest whole number. Any leftover cash stayed in the account.

For example, TSM had a target weight of 29.83%. With a \$100,000 portfolio, the target investment amount was approximately \$29,830. Using a purchase price of \$367.24 per share, the number of shares purchased was:

$29,830 / 367.24 = 81.2$

After rounding down, we bought 80 shares of TSM. We used this same process for all ten stocks. The actual weights ended up a little different from the targets because of rounding and small price changes when we placed the trades.

Step 7 - Rebalance

After reviewing the current holdings, we think our portfolio has performed well, and there have been no major changes that make rebalancing necessary so far. Some weights in the portfolio have changed over time because different stocks performed differently. In our opinion, this

portfolio still remains our original investment idea and is well diversified across industries. Since the timeframe of this project is relatively short and it hasn't been very long from when we bought this portfolio, we believe that making additional trades now may not have a very meaningful impact on the final results.

Therefore, we decided to keep the current holdings and not rebalance the portfolio. The performance analysis in the following step is based on the current portfolio.

Step 8 - Performance Evaluation

Benchmark

We use an equal-weighted portfolio of the initial 20 stocks in the universe from our spreadsheet as our benchmark. The benchmark assumes that each of the 20 selected stocks has the same weight. Then we compare the benchmark to our selected portfolio with scoring, optimizing, and weighting. In this way, we can determine whether our stock portfolio has a better performance than buying all stocks equally.

Comprehensive Dashboard

Performance KPIs

KPI	Portfolio	Benchmark
Total Return / TWR	15.1348%	9.1606%
Money-Weighted Return	15.1348%	9.1606%
Excess Return over Benchmark	5.9742%	—
Annualized Return	77.3271%	42.7973%
Annualized Sharpe Ratio	3.3229	2.9971
Sortino Ratio	5.3840	4.6868

Alpha	8.5073%	0.0000%
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Information Ratio	1.7777	—
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The time-weighted return and money-weighted return are the same because there were no additional cash flows after the initial investment.

KRIs

Risk Metric	Portfolio	Benchmark
Annualized Volatility	23.2710%	14.2794%
Beta vs Benchmark	1.4049	1.0000
Maximum Drawdown	-9.2321%	-7.6313%
Downside Deviation	14.3627%	9.1337%
Worst Daily Return	-3.3203%	-2.5462%
Tracking Error	13.1343%	—

Our portfolio has higher return and better risk-adjusted performance, but it also takes more risk. The higher beta, volatility, and drawdown show that the portfolio is more aggressive than the benchmark.

Overall, our portfolio outperformed the benchmark in both return and risk-adjusted performance, although it also involved higher risk.

Results

From April 20 to May 8, the overall performance of our portfolio is better than the S&P 500's. The cumulative return rate of the portfolio was 5.56%, higher than that of the S&P 500 in the same period. Which means our stock selection and industry configuration have relatively good return performance. Specifically, some sectors strongly pushed the portfolio going up.

From the perspective of daily returns. The daily average earning is about 0.40%, which is 0.11% higher than the S&P 500, showing that our portfolio has stronger upward momentum. However, the risk borne by the combination is also significantly higher. The daily volatility of the portfolio is 1.41%, which is higher than that of the S&P 500 at 0.70%. This indicates our portfolio offers higher returns, the price fluctuations are also greater.

Based on risk-adjusted performance, the Sharpe ratio of the portfolio is 4.47 lower than that of the S&P 500 at 6.65, which indicates that although the portfolio achieved a higher absolute return, the return for each unit of risk taken was not as good as that of the market. Meanwhile, the maximum drawdown of the combination is -1.96%, which is greater than that of the S&P 500 at -0.63%. That means the risk exposure of our portfolio is more obvious than the market.

Conclusion

In conclusion, based on our analysis, our portfolio outperformed the market over most of the investment period, which reflects the effectiveness of our stock selection, industry allocation, and portfolio optimization framework. Overall, the results suggest that combining macro analysis, fundamental analysis, and quantitative optimization can be an effective way to build a diversified portfolio.